Problem Set 3

ECON 40201 - Advanced Industrial Organization II

First Draft (No LLM): March 2nd, 2026
Final Submission: March 8th, 2026

Please typeset your assignment in LaTeX. For the coding part, feel free to use your preferred programming language. Please comment your code clearly, as the clarity of your code will also be evaluated. To the extent possible, all tables, figures, and numerical results in the write-up should be reproducible via a single execution of the submitted code. All answers need to be contained in the write-up. Submit on Canvas your write-ups as a single pdf file and *separately* codes zipped into a single zip file. You can work in a group of two (please clearly indicate both [enrolled not auditing] authors in your submitted PDF) and submit a single file per group. *The first draft will account for 10% of the problem set grade but will only be graded on completion.*

Exercise 1. eBay Auction - Theory and Simulation

Consider a second-price auction where valuations are i.i.d. with $v_i \sim N(0,1)$, and there is no reserve price.

1. What is the equilibrium bid function of the bidders?

Simulate $N = 1,000$ auctions with $i = 1, ..., 10$ bidders each.

2. What are the means of the (i) highest bid and (ii) second highest bid?

Keep the same simulated dataset, but assume the auction is an ascending auction in the style of eBay (see their website for details). Assume that bidders arrive in order of their numbers, that it is their only chance to submit a bid (e.g., bidders 1 and 2 submits their value, creating a new minimum bid $\min\{v_1, v_2\}$ faced by bidder 3), and they do not know which bidder number they are.

3. How does the equilibrium bid function differ from question 1.1 above? Does this change affect either the (i) outcome or (ii) estimation of this auction? If so, how, and if not, why not? You do not need to reference the simulated dataset for this part of the question - just talk about the mechanism for the outcome and what assumptions, if any, change for the estimation relative to the second-price, sealed bid auction.

Using the eBay rules, generate the minimum bid faced by each bidder. Ignore eBay's bid increment rule - any new bidder with a value above the second highest of the previous bids can enter the auction, even if their valuation is only slightly higher. Indicate which bidders will actually submit a bid in each auction.

4. Report the summary statistics of the number of bidders that bid in each auction.

5. Compute the mean of each of the submitted bids for each auction, and the variance of each item's mean. Can you reject that these submitted bids have mean zero?

6. Use Song (2004)'s approach (assume a normal distribution, though more flexible non-parametric approaches are possible) to estimate the bid distribution with the following order statistic pairs of observed bids: (i) first-second highest and (ii) first-third highest. Do you get the same results? Why?

7. Explain how this approach relies on the bid function you described above and the available dataset, and when it might fail. Propose an alternative estimator that does not rely on observing the highest bid in the auction. Is the normality assumption optional, as it was in the previous question? Why?

Exercise 2. eBay Auction - Empirical Exercise

Load the files `bids.csv`, `sparse_attributes.csv`, and `items.csv`. These files are actual bid data from eBay auctions, with the same rules as discussed in section 1 above. The attributes file's columns are the row, column, and value of a sparse matrix (i.e., mostly zeros except where indicated). Make sure to convert this file into a matrix you can use for estimation - keep it as a sparse matrix or else it will be memory-intensive! Each column of the sparse matrix is an indicator function for whether an item's title contained a given word.

1. Report relevant summary statistics for auctions and log-bids.

Assume values take the form $b_{ij} = \exp\left(\gamma\left(X_j\right) + v_{ij}\right)$, where v_{ij} is i.i.d. across bidders.

2. Using the results from section 1, explain how selection affects a regression of $\ln\left(b_{ij}\right)$ on X_j.

3. Denote information sets $\mathcal{I}_1$ as bids from all participants in an auction (whether or not they bid) and $\mathcal{I}_2$ as bids from all bidders. For which bids (if any) does $\mathbb{E}\left[\ln\left(b_{ij}\right) \mid \mathcal{I}_1\right] = \mathbb{E}\left[\ln\left(b_{ij}\right) \mid \mathcal{I}_2\right]$?

4. Suppose you have $\mathcal{I}_3$, which consists of the bids of all bidders and the number of total participants (including non-bidders) in the auction. For which bids (if any) does $\mathbb{E}\left[\ln\left(b_{ij}\right) \mid \mathcal{I}_1\right] = \mathbb{E}\left[\ln\left(b_{ij}\right) \mid \mathcal{I}_3\right]$?

Use the column `pred_n_participant` as the true number of participants (i.e., you now have $\mathcal{I}_3$).

5. Using your reasoning from questions 2.2-2.4, explain how you can use $\mathcal{I}_3$ to estimate γ. Make any assumptions about the distribution of v_{ij} that you wish at this stage (except that it is degenerate), and explain why. (Hint: What distribution should v_{ij} follow so that their maximum (which you observe) follows a well-known distribution with a mean that is related to the number of total participants?)

6. (You may use a high-level package like Scikit-Learn for this question). Use each of the following methods to estimate γ. What is the mean square error from each method?

   (a) Linear regression

   (b) Lasso regression (choose and report your penalty parameter)

   (c) Neural network regression (choose and report your network architecture, including any dropout layers or other features you use)